IINTS-AF Impact, Ethics, And Maintenance Brief

Competition evidence bundle for EUCYS review

Project: IINTS-AF SDK

Purpose: Explain why the SDK matters, how the project stays honest, and what maintenance evidence supports it.

Audience: EUCYS judges, general science judges, ethics reviewers, project mentors.

This PDF was rendered from the markdown report and packaged benchmark artifacts inside the IINTS-AF SDK workspace.

The Problem

Insulin algorithms sit in a high-risk domain. A beautiful graph is not enough evidence. A responsible project must also show how the data was checked, which baselines were used, what failed, how edge cases behaved, and whether the results can be reproduced.

Many student projects stop at a demo. IINTS-AF aims to go further: it creates the scaffolding needed to evaluate insulin decision systems in a reviewable way.

Why This Matters

Stakeholder	Why the SDK matters
Students and researchers	They can test algorithms without hiding logic in notebooks
Mentors and judges	They can inspect commands, outputs, and limitations
Open-source community	They can reproduce, critique, and improve the work
Patients indirectly	Safer research practices reduce overclaiming before real-world use

The most important impact is methodological: making pre-clinical algorithm research less opaque.

The Safety Philosophy

The project follows a conservative safety philosophy:

- separate algorithm output from safety supervision
- log interventions instead of hiding them
- evaluate corrupted data instead of assuming perfect inputs
- report low-glucose risk, not just average performance
- state limitations clearly
- never present the SDK as a medical device

This is important because software confidence should come from evidence, not from optimism.

Ethics Boundary

IINTS-AF is for simulation, research, education, and reproducible benchmarking.

It is not for:

- real patient dosing
- replacing a clinician
- controlling an insulin pump in daily life
- making individual treatment recommendations
- claiming clinical efficacy without clinical evidence

That boundary should stay visible in the repository, the documentation, and all EUCYS materials.

What Makes The Project Responsible

Practice	Why it is responsible
Fixed seeds	Reduces cherry-picking risk
Baselines	Prevents comparing only against nothing
Corrupted-data arm	Shows sensitivity to bad inputs
Supervisor-off ablation	Shows what the safety layer changes
Limitations section	Prevents overclaiming
Source legend	Makes scientific assumptions traceable
CI checks	Shows the project is maintained, not frozen

Maintenance Evidence

After the final evidence refresh, the project passed the main local checks:

Check	Result
Full Python test suite	369 passed, 4 skipped
Flake8	passed
Mypy	passed
MkDocs build	passed
MDMP sync check	passed
Git diff whitespace check	passed

Remote GitHub Actions for the latest pushed evidence commit also completed successfully for Python Package CI, Health Badges, and Notebook Bake.

Why Open Source Is Part Of The Science

Open source matters here because insulin-algorithm claims should be inspectable. A closed demo can look convincing while hiding assumptions. An open SDK can be challenged.

A judge can ask:

- Which command produced this result?
- Which seed set was used?
- Which baselines were compared?
- What happens when data is corrupted?

- What does the safety layer change?
- Which files prove the benchmark design?

IINTS-AF is designed so those questions have file-based answers.

Honest Result Framing

The final benchmark is not presented as "my algorithm solves diabetes." That would be wrong.

The result is framed as:

- the SDK can run a large reproducible benchmark
- clean certified data performed better than corrupted uncertified data
- the candidate performed strongly in the clean arm
- another baseline had the highest full-bundle TIR
- the safety supervisor is a trade-off layer that must be measured, not blindly praised

This honest framing is stronger for EUCYS because it shows scientific maturity.

Future Impact Roadmap

The best next improvements are:

Next step	Why it helps
Real-data realism dashboard	Shows whether simulated curves resemble real CGM behavior
More public dataset adapters	Makes validation less dependent on one data source
Failure-case gallery	Turns edge cases into inspectable learning artifacts
Calibration analysis	Shows whether predictor uncertainty matches actual error
Reproducible Docker/Devcontainer	Makes judging and review easier on other machines
Smaller public sample bundle	Lets reviewers run a fast smoke benchmark

EUCYS-Level Contribution

The EUCYS contribution is interdisciplinary:

Discipline	Contribution
Computer science	SDK architecture, CLI, tests, reports, reproducibility
Data science	data validation, metrics, uncertainty and benchmark design
Biomedical engineering	insulin-control simulation and safety-supervisor framing
Ethics	no medical overclaiming, transparent limitations, reproducible evidence

Core Impact Claim

The strongest impact claim is:

IINTS-AF makes insulin-algorithm research easier to test, easier to criticize, and harder to overclaim.